MZUZU UNIVERSITY

Department of Mathematics

EMT 1101 - Precalculus

Continuous Assessment Examination II

Sunday, March 19, 2017 **Time:** $14:00 - 16:00$ **(2 hours)**

1. Check that the paper has **questions**.
2. Attempt all questions
3. Show all your working clearly
4. Use proper notation
5. Simplify your answers as far as possible
6. **No Calculators**

1. (a) Let $h(x) = x^2 - 4x - c$. Find a nonzero value c such that $h(c) = c$. (3)

Solution: $h(c) = c$ implies
$c^2 - 4c - c = c$
$c^2 - 6c = 0$
$c(c - 6) = 0$
$c = 0$ or $c = 6$. Hence the nonzero value is 6

(b) Let $f(x) = \dfrac{x - a}{x + a}$. Show that $f(5a) = 2f(2a)$

Solution: $f(5a) = \frac{5a-a}{5a+a} = \frac{4a}{6a} = \frac{2}{3}$.
$2f(2a) = 2\frac{2a-a}{2a+a} = 2\frac{a}{3a} = 2\frac{1}{3} = \frac{2}{3}$.
Hence $f(5a) = 2f(2a)$

2. Let $M(x) = \frac{(2x-1)}{(x-2)}$.

(a) Compute $M[M(x)]$ (3)

Solution: $M(M(x)) = M(\frac{2x-1}{x-2}) = \frac{2(\frac{2x-1}{x-2})-1}{(\frac{2x-1}{x-2})-2}$
$= \frac{4x-2-x+2}{2x-1-2x+4} = x$

(b) Let $g(x) = 2x + 1$. (3)
Find $f(x)$, given that $(g \circ f)(x) = 10x - 7$

Solution: $g(f(x)) = 10x-7$, and similary, $g(f(x)) = 2f(x)+1$ so $10x-7 = 2f(x) + 1$ then $f(x) = 5x - 4$

3. (a) Let $f(x) = 4x^2$, find

$$\frac{f(x) - f(a)}{x - a}$$

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(b) Suppose that the demand function for a certain item is given by
$p(x) = \frac{24}{2^{\frac{x}{100}}}$, where x is the number of items that can be sold when the price of each item is $p(x)$ kwacha.
Compute $\frac{\Delta p}{\Delta x}$ over the interval $0 \leq x \leq 100$

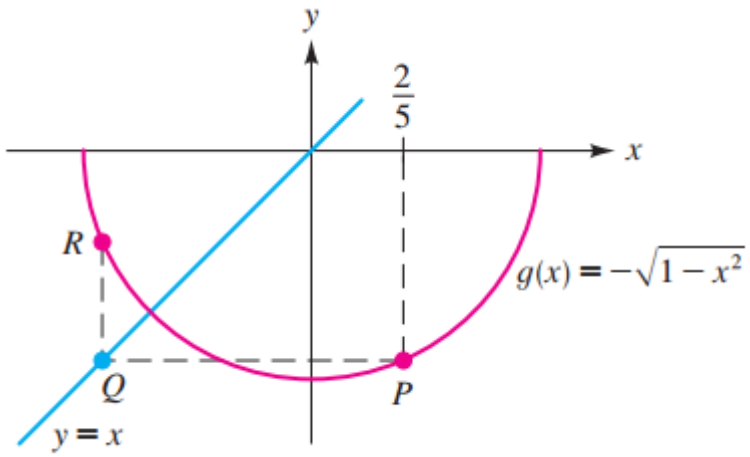
4. Sketch the graph $y = e^{-x} - e$ and state the asymptote, $x-$ and $y-$intercepts, domain and range. (5)

Solution: the graph of $y = e^x$ reflected on $y - axis$ then translated downwards e units asymptote$= -(2 - e)$, $x-$intercept $= -1$, $y-$ intercept $= -(1 - e)$, domain;(∞, ∞), Range;$(-(1 - e), \infty)$

5. Sketch a graph of the function and state the domain and range

$$A(x) = \begin{cases} x^3, & -2 \le x \le -1 \\ x^2, & x > -1 \end{cases}$$

6. Determine the coordinates of the points P,Q and R.



7. (a) Estimate 4^{50} in terms of powers of 10.

Solution: Use the formula $2^{10} \approx 10^3$
$(2^2)^{50} = 2^{100} = (2^{10})^{10} \approx (10^3)^{10} = 10^{30}$

(b) Use properties of exponents to simplify

$$\left(\sqrt{5^{\sqrt{2}}}\right)^2$$

and write the answer in form of b^n where b and n are real numbers.

Solution: $5^{\sqrt{2}} = 5^{2^{\frac{1}{2}}}$

(c) Simplify $(\log_2 3)(\log_3 4)(\log_4 5)$ (3)

Solution: Using change of log bases we have $(\frac{\log 3}{\log 2})(\frac{\log 4}{\log 3})(\frac{\log 5}{\log 4}) = \frac{\log 5}{\log 2} = \log_2 5$

8. Let $f(x) = 2^x$ and let $g(x)$ denote the function that is inverse to $f(x)$. On the same set of axis sketch the graphs $y = x$, $f(x)$ and $g(x)$.

Solution:

9. Let $f(x) = x^3 + 2x + 1$ and assume $f^{-1}(x)$ exist and has domain $(-\infty, \infty)$. Simplify each expression.

(a) $f(f^{-1}(4))$

Solution: 4

(b) $(f \circ f^{-1})(\sqrt{2})$

Solution: $(f \circ f^{-1})\sqrt{2} = f(f^{-1}(\sqrt{2})) = \sqrt{2}$

10. Let $f(x) = \frac{2x-3}{x+4}$. Find f^{-1}.

Solution: Let $y = \frac{xx-3}{x+4}$. Interchange x and y and then solve for y $x = \frac{2y-3}{y+4}$ and $f^{-1}(x) = y = \frac{-3-4x}{x-2}$

11. (a) Determine the domain of $h(t) = \sqrt{\frac{3t-4}{7-2t}}$ present it using interval notation.

Solution: We want t for which $\frac{3t-4}{7-2t} \geq 0$. The key numbers are $t = \frac{4}{3}$ and $t = \frac{7}{2}$ which yields the intervals $(-\infty, \frac{4}{3})$, $(\frac{4}{3}, \frac{7}{2})$ and $(\frac{7}{2}, \infty)$

interval	test value	$3t-4$	$7-2t$	$\frac{3t-4}{7-2t}$
$(-\infty, \frac{4}{3})$	0	-	+	-
$(\frac{4}{3}, \frac{7}{2})$	2	+	+	+
$(\frac{7}{2}, \infty)$	4	+	-	-

Hence $\frac{3t-4}{7-2t} \geq 0$ has solution set $\left[\frac{4}{3}, \frac{7}{2}\right]$ which yields the domain

(b) Determine the range of $g(x) = \frac{4x-20}{3x-18}$ and present it using interval notation.

Solution: Let $y = \frac{4x-20}{3x-18}$. Then
$3xy - 18y = 4x - 20$
$3xy - 4x = 18y - 20$
$x(3y-4) = 18y - 20$
$x = \frac{18y-20}{3y-4}$
Hence the range comprises all real numbers y such that $3y - 4 \neq 0$, i.e. $(-\infty, \frac{4}{3}) \cup (\frac{4}{3}, \infty)$

12. (a) Evaluate $\log_9 27$.

Solution: Let $y = \log_9 27$, then $9^y = 27$
$(3^{2y} = 3^3$
$2y = 3$
$y = \frac{3}{2}$

(b) Find the domain of $y = \log_5(3 - 4x)$ and present it in interval notation.

Solution: The domain consists of all real numbers x s.th. $3 - 4x > 0$
$4x < 3$
$x < \frac{3}{4}$
Hence the domain is $(-\infty, \frac{3}{4})$

13. (a) Solve each equation $3^z = 9\sqrt{3}$

Solution:

$$\begin{aligned} 3^z &= 3^2 3^{\frac{1}{2}} \\ 3^z &= 3^{\frac{5}{2}} \\ z &= \frac{5}{2} \end{aligned}$$

(b) Solve the equation $\log_2(x+4) = 2 - \log_2(x+1)$ for x.

Solution: $\log_2(x+4) = 2 - \log_2(x+1)$
$\log_2(x+4) + \log_2(x+1) = 2$
$\log_2(x+4)(x+1) = 2$
$x^2 + 5x + 4 = 4$
$x^2 + 5x = 0$
$x(x+5) = 0$
$x = 0$ or $x = -5$
Upon checking $x = 0$ is the only solution

(c) Solve the inequality $4^{5-x} > 15$, presenting your answer in interval notation

Solution: Taking natural log of either side yields
$\ln 4^{5-x} > \ln 15$
$(5-x)\ln 4 > \ln 15$
$5 - x > \frac{\ln 15}{\ln 4}$
$x < 5 - \frac{\ln 15}{\ln 4}$
Hence the solution set is $(-\infty, 5 - \frac{\ln 15}{\ln 4})$